End Sem Exam 27th April 2026

Optical Remote Sensing

Time: 180 minutes ***Total Marks: 80***

Note: The right column shows the Marks for each set of questions.
If need, make suitable assumptions and state it clearly when answering

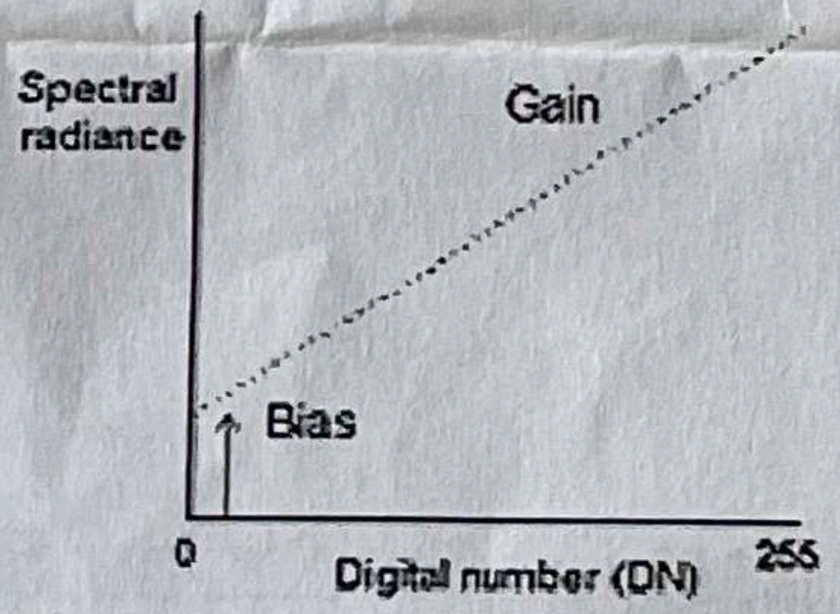
Part- I: [10 x 1 = 10]

1. What is the basic difference between contrast and spatial enhancement?
2. List the factors considered for PFZ mapping?
3. If Venus and Mars both have atmospheres dominated by CO_2, why is Venus extremely hot while Mars is very cold?
4. With respect to image analysis, what is the advantage of RNN over CNN?
5. Differentiate between DSM and DTM.
6. In IPCC standards, how are emission quantified in 'bottom-up' approach?
7. Define Watershed?
8. Why is CO_2 used as the standard reference gas for measuring Global Warming Potential of other greenhouse gases?
9. What approach is used to retrieve Aerosol Optical Depth from satellite data?
10. What colour does land degraded by wind and water erosion typically appear in satellite images?

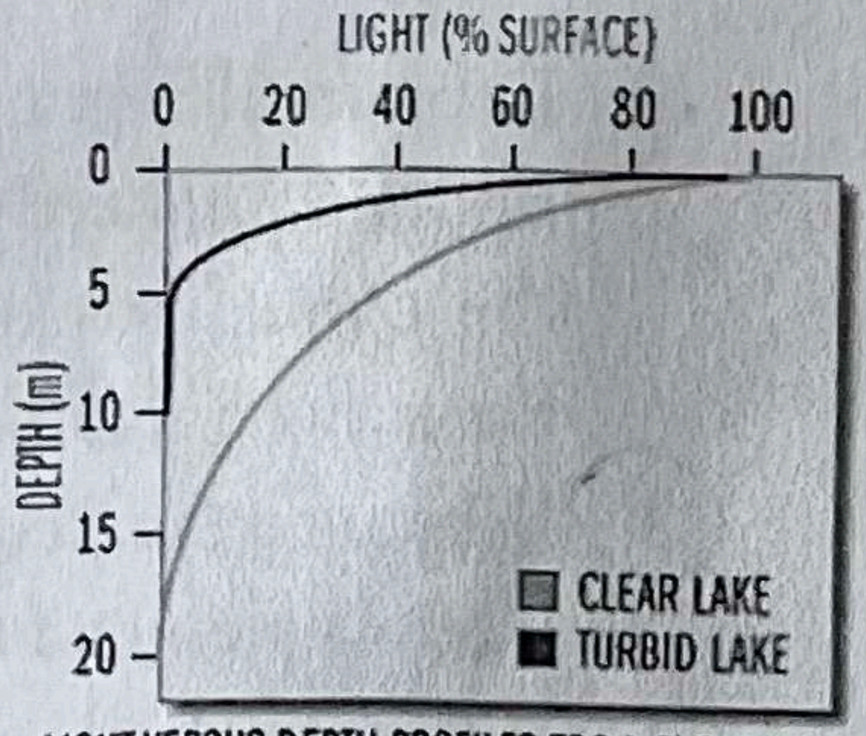
grey / yellow

bright

Part-II: [6 x 2 = 12 Marks]



1. From the given DN–radiance graph:
   (a) Derive the equation used to convert Digital Number (DN) to spectral radiance.
   (b) Clearly define each parameter in the equation.
2. According to the Beer–Lambert law, attenuation depends on concentration and one more factor. Identify this factor and explain its role in absorption.
3. How would you differentiate following in a satellite image using spectral reflectance characteristics (a) Floating NIR plastic and water surfaces (b) Snow and clouds PCA
4. Based on the given figure (light vs depth), define the concept of light attenuation in water as discussed in class.
5. List the four methods (sources) used for DEM generation. ESSA
6. Define the parameters in the equation: $\rho = \pi (L_T - L_p) / (E \times T)$.



LIGHT VERSUS DEPTH PROFILES FOR A CLEAR LAKE

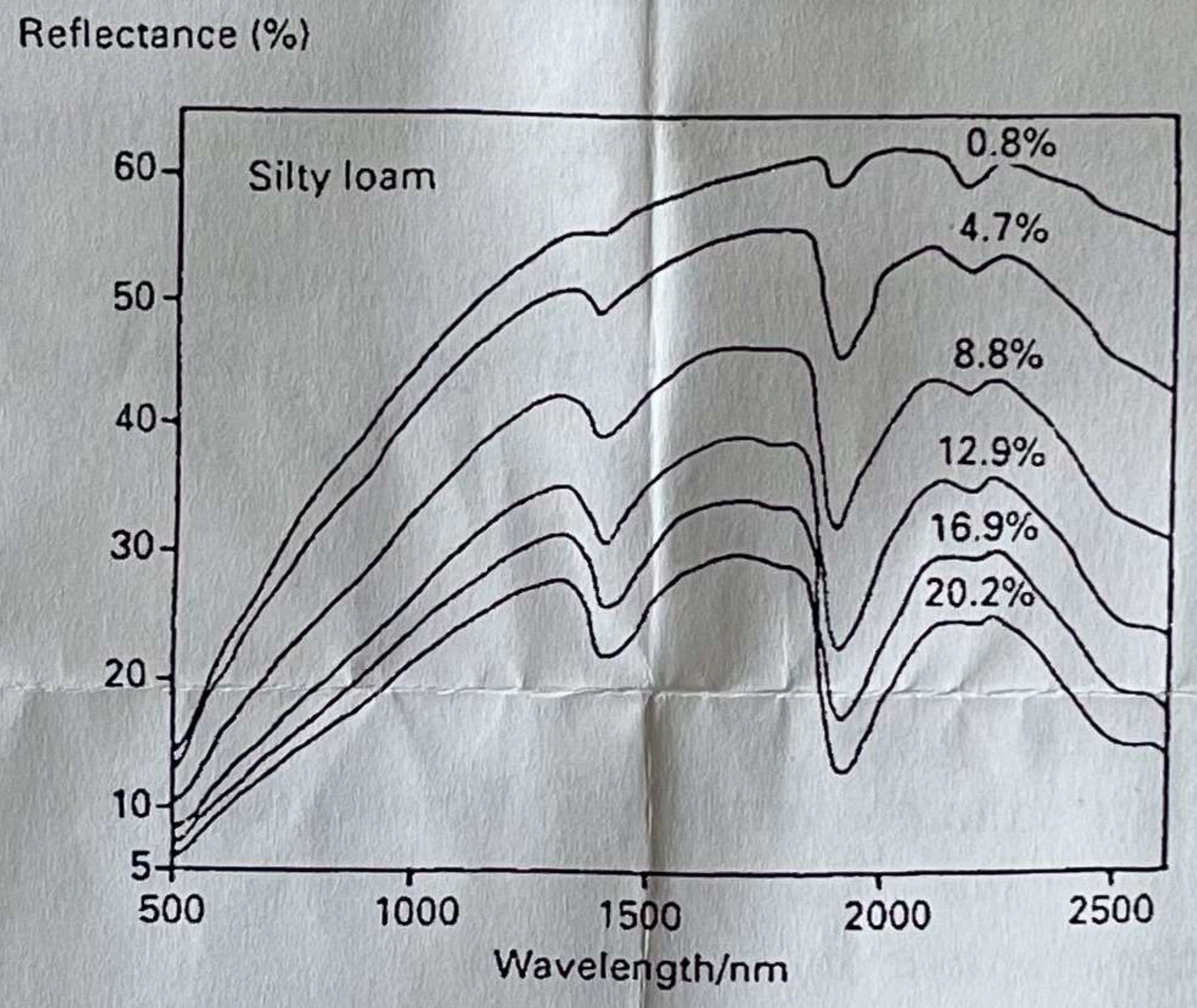
Part-III: [6 x 3=18 Marks]

1. What is the difference between infrared (IR) absorption spectroscopy and emission spectroscopy when retrieving GHGs? Give examples of any one GHG per method with specific wavelengths.
2. Explain the environmental conditions leading to phytoplankton bloom formation in the northern Arabian Sea?

3. Why is per-pixel classification less suitable for very high resolution images? (6 points)
4. In an interview you are asked to define certain situations where you will use (i) satellite imagery and aerial photographs (ii) optical and microwave data (iii) multi-spectral and hyper spectral images. Elucidate your answers with examples effectively?
5. Match the following

Image Arithmetic	Applications
A. Image multiplication	1. Site suitability analysis C
B. Spatial averaging	2. DN to radiance conversion D
C. Image ratio	3. Illumination effect reduction F
D. Weighted linear combination	4. Change detection (urban expansion) E
E. Image subtraction	5. Masking operation A
F. Image scaling	6. Noise reduction B

6. Identify and explain any three key features shown in the following figure



Part-IV: Answer any SIX [6 x 5 = 30 Marks]

1. Explain El-Nino southern oscillation and how it impacts Indian rainfall?
2. Explain the need for solar irradiance correction in satellite imagery preprocessing. Derive the correction for (a) Sun elevation angle and (b) Earth–Sun distance, and show how these two factors are combined into a single radiometric correction equation
3. What are the distinct morphological features of urban sprawl? Explain how agent-based models simulate and map its spatial evolution?
4. When estimating components of the surface energy balance (like albedo, net radiation, and heat fluxes), why can 3D surfaces derived from Photogrammetry lead to different energy balance estimates compared to LiDAR, even if both produce similar geometric accuracy? Discuss in terms of radiative geometry and surface representation.
5. Write the radiative transfer equation governing the transfer of solar radiation from the Sun through the atmosphere to the ocean surface. Define all parameters

6. Discuss how Principal Component Analysis is applied in: (a) change detection studies. (b) image fusion.
7. Explain (a) the concept of fuzzy classification in satellite image analysis. How does it differ from traditional (hard) classification, and why is it more suitable for mixed pixels?
   (b) A pixel has a DN value of 80. Consider two land cover classes with the following statistics:

Class	**Mean (μ)**	**Standard Deviation (σ)**
Forest	75	10
Urban	110	15

   Using the Gaussian membership function: Calculate the membership values for both classes.

$$\mu_i(x) = \exp\left(-\frac{(x-\mu_i)^2}{2\sigma_i^2}\right)$$

<u>Part-V: Answer any ONE [1 x 10 = 10 Marks]</u>

1. Explain Image segmentation process used in object based image classification?
2. What is the approach followed in 'bottom-up' approach of quantifying GHGs from fires? Explain the parameters and also state which parameters are derived using (a) remote sensing data, (b) ground measurements and (c) both